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23 May 1978
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20 March 1972

MILITARY SPECIFICATION

COVERALLS, SUBMARINE DECK EXPOSURE

This specification is approved for use by the Naval Supply Systems Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for a hooded, submarine deck exposure coverall constructed with a coated nylon fabric outershell, a plastic foam interlining, and a nylon fabric lining.

1.2 Classification.- The coverall shall be of one type only, and in the following sizes (see 6.2):

Schedule of sizes

Small (34-36)	Large (42-44)
Medium (38-40)	X-Large (46-48)

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-F-106	- Fasteners, Slide Interlocking
V-T-295	- Thread, Nylon
JJ-W-155	- Webbing, Textile, (Cotton, Elastic)
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)
DDD-B-1199	- Binding, Textile, Cotton, Bias-Cut

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

MILITARY

MIL-P-12420 - Plastic Material Cellular, Elastomeric
MIL-B-17757 - Box, Shipping, Fiberboard (Modular Sizes)
MIL-C-19759 - Cloth, Coated, Twill, Nylon (Low Count)
MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic
MIL-C-21852 - Cloth, Nylon, Taffeta (2.0 Ounce)
MIL-C-43256 - Cord, Polyester, Solid Braid
MIL-C-43836 - Cloth, Interlining, Non-Woven

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Guide sample.- Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample in which case this specification shall govern.

3.2 First article.- When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval.

3.3 Materials.-

* 3.3.1 Basic Material.- The basic material for the outer shell shall be chloroprene coated nylon conforming to Type I of MIL-C-19759, except that the uncoated side of the finished (coated) cloth shall match Shade Orange 3604. The coating compound shall be pigmented orange. In addition, tear strength in the filling direction shall be a minimum of 3400 grams, hydrostatic resistance shall be after 40 hours exposure in the weatherometer and the flex stiffness shall be tested at -20°F.

* 3.3.2 Lining.- The material for the coverall lining shall be nylon taffeta conforming to MIL-C-21852 except that the color may be olive green, black or approximately the same shade as the basic material.

* 3.3.3 Interlining.- The material for the interlining shall be plastic foam with a thickness of 1/8 inch (3.2mm) and 1/4 inch (6.4mm) conforming to type II, class 6 of MIL-P-12420, except that the buoyancy shall not be less than 54.0 pounds per cubic foot when tested as specified in 4.4.1.

* 3.3.4 Thread.- The thread for seaming and stitching the coverall shall be nylon, conforming to type I, class 1, subclass B, sizes B and AA of V-T-295. The color shall be orange TH, Cable No. 66074.

3.3.4.1 Colorfastness.- The dyed thread shall show colorfastness to weathering and laundering equal to or better than the standard sample when tested as specified in V-T-295. When no standard sample is available, the dyed thread shall show a minimum of "good" fastness to weathering and laundering when tested as specified in V-T-295.

* 3.3.5 Slide fasteners.-

3.3.5.1 Utility pocket closure.- The slide fastener shall be 5 inches (12.7cm) long, natural brass or bronze conforming to type I, style 7, size L, short tab pull of V-F-106. The color of the tape shall be black or orange and shall show good colorfastness to laundering (see 4.4.1).

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3.3.5.2 Front closure.- The slide fastener Shall conform to type I, style 3, size MS, short tab pull, natural brass or bronze chain, conforming to V-F-106. All metal components of the slide fastener shall be furnished by the same supplier. The length of the slide fastener shall be such that the top stop of the slide fastener shall be set not more than 1/4 inch (6.4mm) from the finished neck seam with the bottom stop of the slide fastener set as indicated on patterns.

3.3.5.2.1 The tape for the slide fastener shall be cotton, synthetic or a combination of both. The color of the tape shall be black or orange and shall show good colorfastness to laundering. The thong shall be a nylon braid or webbing approximately 3/8 inch (9.5mm) in width with a finished hanging length of 4 1/2 to 5 1/2 inches (11.4 to 14.0cm). The color of the thong shall be olive green, black or orange and shall show good colorfastness to laundering (see 4.4.1).

3.3.6 Fastener tapes.- The matching hook and pile fastener tapes shall be 1 inch (2.5cm) wide for the waist adjustment straps and 2 inches (5.0cm) wide for the sleeve and leg adjustments, black or orange in color, conforming to type II, class 3 of MIL-F-21840.

* 3.3.7 Cord.- The polyester cord for the hood adjustment drawcord shall be 3/16 inch (4.8mm) diameter, natural, olive green or black conforming to MIL-C-43256, except that when dyed it shall also show good colorfastness to crocking (see 4.4.1).

3.3.8 Identification and instruction label.- Each coverall shall have a combination identification, size, and instruction label conforming to type I, classes 1, 2, and 3 of DDD-L-20. The printing shall show fastness to laundering. The labels shall bear the following inscription:

COVERALL, SUBMARINE DECK EXPOSURE
CONTRACT NO: DSA-100-00-0-0000 (Example)
STOCK NO: 8415-00-000-0000 (Example)
SIZE: Medium 38-40 (Example)
NAME OF SUPPLIER:

DO NOT DRY CLEAN

Hand laundering with warm water and soap is preferred. May be laundered below 140°F and tumble dried below 160°F. After each use, apply silicone or petroleum jelly to slide fastener chain.

3.3.9 Elastic webbing.- The elastic webbing for the waistband adjustment strap shall be cotton conforming to type I, class 3, 3/4 inch (19.0mm) wide of JJ-W-155.

* 3.3.10 Cement.- The cement used for adhering the abutted seams of the foam interlining shall have a chloroprene rubber base. The use of reclaimed rubber is prohibited. The abutted seams shall have a seam strength of 10 pounds per 2 inch width (1750 N/m) when tested as specified in 4.4.1.2.

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3.3.11 Tape.- The cotton material used for extending the foam interlining shall conform to type II, class 1 of DDD-B-1199.

* 3.3.12 Pocket welt interlining.- The pocket welt interlining shall be a non-woven fabric conforming to type II, class 1 of MIL-C-43836.

* 3.4 Design.- The impermeable coveralls shall have an outershell of coated nylon fabric with the coating finishing on the underside. The coverall shall be lined with an uncoated nylon taffeta and interlined with a layer of expanded plastic foam. The upper front and back sections shall be insulated with 1/4 inch (6.4mm) thick foam. Remainder of coverall shall have 1/8 inch (3.2mm) thick foam. The coverall shall have an attached hood with drawcord. The sleeve and leg bottoms and waistband shall have hook and pile tape fastener adjustments. Coverall shall have a welted open pocket at each hip and a zippered utility pocket on the left sleeve. Front closure shall have a slide fastener from neck to crotch with a protective flap.

3.4.1 Figure.- Figure 1 is furnished for information purposes only. When inconsistences exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns.- Standard patterns used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow 3/8 to 1/2 inch (9.5 to 12.7mm) for all seams, except as otherwise indicated in Table I. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

* 3.5.1 Pattern parts.- The component parts shall be cut from the material specified in accordance with the following pattern parts:

Material	Pattern Parts	Cut Parts
Cloth, coated, Twill, nylon (outershell)	Back upper	1
	Back trouser	2
	Front	2
	Top sleeve	2
	Undersleeve	2
	Front facing	2
	Back waist channel	1
	Pocket (inner)	2
	Pocket (outer)	2
	Long utility pocket (left sleeve)	1
	Short utility pocket (left sleeve)	1
	Narrow accessory pocket (left sleeve)	1
	Wide accessory pocket (left sleeve)	1
	Pocket welt (bottom ply)	2
	Pocket welt (top ply)	2
	Adjustment tab (sleeve bottom)	2 1/
	Adjustment tab (leg bottom)	2 1/
	Hood, side panel	2
	Hood, center panel	1
	Hood, cord channel	1
	Adjustment tabs (waistband)	1 1/
	Protective flap	1

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Material	Pattern Parts	Cut Parts
Plastic, foam, elastomeric (interlining)	Back, upper - 1/8 inch	1
	Front and back yoke - 1/4 inch	2
	Back, upper trouser - 1/8 inch	2
	Back, lower trouser - 1/8 inch	2
	Center front trouser - 1/8 inch	2
	Side front, trouser - 1/8 inch	2
	Top sleeve - 1/8 inch	2
	Undersleeve - 1/8 inch	2
	Hood, side panel - 1/8 inch	2
	Hood, center panel - 1/8 inch	1
	Protective flap - 1/8 inch	1
Cloth, nylon, taffeta (lining)	Back, upper	1 2/
	Back trouser	2
	Front	2
	Top sleeve	2
	Undersleeve	2
	Hood, side panel	2
	Hood, center panel	1
Cloth, non-woven	Pocket welt interlining (bottom ply)	2
	Pocket welt interlining (top ply)	2

1/ Adjustment tabs may be stitched in strip form and cut to meet the finished dimensions of pattern parts.
2/ This part may be cut in two pieces with seam allowance added.

3.6 Construction.-

3.6.1 Stitch, seam, and stitching types.- Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seams or stitches are given for the same part of the operation, either one may be used. Where stitch type 401 is used, the looper or underthread shall be on the inside of the coverall. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats or open seams will result. Topstitching and edgestitching, unless otherwise specified, shall be 1/16 inch (1.6mm) from the edge. Double needle stitching shall be 3/16 to 5/16 inch (4.8 to 7.9mm) gauge.

3.6.2 Thread breaks and ends of seams.- The ends of seams and stitching produced with a 301 stitch type, when not caught in other seams and stitching, shall be backtacked not less than 3/8 inch (9.5mm). Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch (12.7mm). Skipped stitches or thread breaks on 401 seaming may be repaired using 301 stitch type. Stitch type 401 shall not be used unless the ends are caught in other seams or stitchings.

3.6.3 Stitches per inch.- The minimum and maximum number of stitches per inch shall be as specified in Table I.

* 3.7 Manufacturing operations requirements.- The coveralls shall be manufactured in accordance with all operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations as specified.

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Table I - Construction of coveralls

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting.-					
	a. Cut the coveralls in strict accordance with patterns which show size, directional lines, and notches for the proper assembling of all parts.					
	b. The face side of the coated nylon shall be the uncoated side and shall be the exposed surface of the coverall.					
	c. Cut all component coated nylon parts out of one piece of material except adjustment tabs, facing, and inner and outer pockets which may be cut from ends. When so cut, the parts shall closely match the main assembly. Drill holes are prohibited.					
*	d. Directional lines on the fronts and back of outer-shell material may be swung from the warp direction by not more 1 1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the warp directional lines to the selvage edge (on the even side of the lay) of the fabric and the difference between the two measurements shall not exceed 1 1/2 inches.					
	e. All component parts of the uncoated nylon taffeta shall be cut from one piece of material, except the sleeves, which may be cut from ends. When cut from ends the shade of the sleeve linings shall be an approximate match.					
	f. The component parts of the plastic foam need not match each other in shade and need not be cut from one piece of material.					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
2.	Replacement of damaged parts.- Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages specified in Section 4 are excluded and replaced with non-defective and properly matched material.					
3.	Size and shade marking.- a. All component parts of the coated fabric outershell and nylon lining shall be marked or ticketed to assure a uniform shade and size assembly throughout the coverall. Component parts which are allowed to be cut from ends need not be marked for shade, but shall be an approximate match to each other.					
	b. Any method of marking may be used except: 1. Metal fastening devices 2. Sew-on tickets. 3. Adhesive type tickets whose adhesive causes discoloration or whose adhesive adheres to the material upon removal of the tickets.					
	d. The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering is legible and does not show on the outside of the coverall.					
4.	Make interlining (plastic foam).- a. Cement (abut) all foam interlining parts to form the interlining shell. Insure that all seams to be cemented are free of foreign matter prior to joining.					
	b. Stitch a continuous length of 1 inch (2.5cm) wide bias-cut tape to the shell at the front opening over-	301 or 401	LSa-1	6-9	IB IB	IB IAA

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Make interlining (plastic foam).- (cont'd) lapping 5/16 to 3/8 inch (7.9 to 9.5mm). Begin at the crotch and continue up and around the hood and down again to the crotch opening. The stitchline shall be 3/16 to 1/4 inch (4.8 to 6.4mm) from the edge of the interlining shell.					
5.	Make lining (nylon taffeta).-					
	a. Stitch dart in each hood side panel, as indicated by marks on pattern, tapering and backtacking each pointed end of dart.	301	OSf-1	8-12	IB	IB
	b. (1) Stitch, face to face, hood center panel to hood side panels.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seams.	503 or 504	SSa-1	8-12	IB	IB
	c. Stitch four (4) darts in trouser portion of fronts and four (4) darts in trouser portion of backs, as indicated by marks on pattern, tapering and backtacking each pointed end of darts.	301	OSF-1	8-12	IB	IB
	d. (1) Superimpose coated side of front facings on face side of front linings. With front edges of facings and linings even, stitch all outer edges of facings 1/8 to 3/16 inch (3.2 to 4.8mm) from raw edges.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge front edge of facings and linings.	503 or 504	SSa-1	8-12	IB	IB
	e. (1) Stitch trouser backs, face to face, at seat.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam.	503 or 504	SSa-1	8-12	IB	IB

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Make lining (nylon taffeta).- (cont'd)					
	f. (1) For two-piece upper back, stitch center back seam, face to face.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam.	503 or 504	SSa-1	8-12	IB	IB
	g. (1) Join upper back to trouser backs at the waist, face to face, forming a pleat on the upper back at the center.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam.	503 or 504	SSa-1	8-12	IB	IB
	h. (1) Join the back to the fronts, face to face, at the shoulders.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seams.	503 or 504	SSa-1	8-12	IB	IB
	i. (1) Join topsleeve and undersleeve, face to face, at sleeve outseam.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam.	503 or 504	SSa-1	8-12	IB	IB
	j. (1) Join sleeves to body, face to face, at armholes.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seams.	503 or 504	SSa-1	8-12	IB	IB
	k. (1) Stitch sleeve inseam and body sideseam, face to face, in one continuous operation.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seams.	503 or 504	SSa-1	8-12	IB	IB
	l. (1) Join fronts at crotch, face to face.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seams.	503 or 504	SSa-1	8-12	IB	IB

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Make lining (nylon taffeta) (cont'd)					
	m. (1) Stitch leg inseams, face to face, in one continuous operation.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam.	503 or 504	SSa-1	8-12	IB	IB
	n. (1) Join the hood to the body at neck, face to face, forming a pleat in the center of the upper back at neckline, simultaneously catching the top edge of the identification (centered) in the row of stitching.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	(2) Overedge seam	503 or 504	SSa-1	8-12	IB	IB
	NOTE: Operations 5.b, e, f, g, h, i, j, k, l, m and n may be performed in one operation by simultaneously stitching and overedging.	515 or 516 or 517 or 518	SSa-2	8-12	IB IB IB IB	IAA IAA IB IB
*6.	Join foam interlining to taffeta lining.-					
	a. Join the foam interlining to taffeta lining at bottom of front closure (crotch), front edge of closure and hood facing in one continuous operation. Position the cotton tape (already attached to foam interlining) on the front edge of prestitched coated fabric facing and taffeta lining. The raw edge of cotton tape shall extend 1/4 to 5/16 inch (6.4 to 7.9mm) beyond the fold line on facing, as indicated by notches on pattern. Stitch tape, facing, and lining together 1/8 to 3/16 inch (3.2 to 4.8mm) from the raw edge of tape.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	NOTE: When the tape is sewn to hood opening, the raw edge shall be even, smooth, and flat.					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
*6.	Join foam interlining to taffeta lining.- (cont'd)					
	b. Turn foam interlining and taffeta lining to finished position. The legs and sleeves of taffeta lining should be inserted into the legs and sleeves of foam interlining. Stitch lining and interlining together at bottom of legs and sleeves approximately 1/4 inch (6.4mm) from raw edge.	301 or 401	SSa-1	8-12	IB IB	IB IAA
*7.	Make accessory/utility pocket.-					
	a. Hem tops of the long and short utility pockets in accordance with notches, stitching 3/16 to 5/16 inch (4.8 to 7.9mm) from folded edge.	301 or 401	EFa-1	8-12	IB IB	IB IAA
	b. Place short utility pocket over long utility pocket with the bottom edges even and join with a single row of stitching down the center and length of short pocket thereby providing two openings for the short utility pocket.	301 or 401	SSv-1	8-12	IB IB	IB IAA
	c. Fold under the bottom and sides of the utility pocket 3/16 to 1/4 inch (4.8 to 6.4mm) and stitch to the accessory pocket as indicated by marks on pattern, approximately 1/16 inch (1.6mm) from folded edge. Reinforce with a second row of stitching parallel to the first row and approximately 1/8 inch (3.2mm) from folded edge.	301 or 401	LSd-2	8-12	IB IB	IB IAA
	d. Bartack the tops of all utility compartment openings, superimposed on the inner row of double stitching at corners and on the center row of single stitching. Bartacks shall be 1/4 to 3/8 inch(6.4 to 9.5mm) long.	301	Bartack	10-16 per bartack	IB	IB

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
*7.	Make accessory/utility pocket.- (cont'd)					
	e. Form a 3/4 inch (19.0mm) bellows by joining the two bottom corners of the wide accessory pocket with a single row of stitching 1/4 inch (6.4mm) from the raw edge.	301	SSa-1	8-12	IB	IB
	f. Stitch, face to face, wide accessory pocket to narrow accessory pocket at top and bottom 7/16 to 9/16 inch (11.1 to 14.3mm) from raw edges leaving an opening for the slide fastener. Fastener shall be centered on length of seamline.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	g. Join accessory pocket and slide fastener together by stitching around fastener opening 1/4 to 5/16 inch (6.4 to 7.9mm) from folded side edges and from top and bottom stops of fastener. The slide fastener pull shall finish at the top of pocket when fastener is closed.	301 or 401	LSd-1	8-12	IB IB	IB IAA
	h. Die cut tow 1/4 inch (6.4mm) drainage holes as indicated by marks on pattern.					
8.	Make sleeve and leg adjustment tabs.-					
	a. Cut a 2 1/8 to 2 3/8 inch (5.4 to 6.0cm) length of 2 inch (5.1cm) wide hook fastener tape for the sleeve adjustment tab.					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Make sleeve and leg adjustment tabs.- (cont'd) b. Cut a 3 3/8 to 3 5/8 inch (8.6 to 9.2cm) length of 2 inch (5.1cm) wide fastener tape for the leg adjustment tab.					
	c. Fold, back to back and abut lengthwise edges of sleeve or leg adjustment tab (as applicable) at center of tab in accordance with fold lines on pattern. Fold lengthwise edges of tab in half again inwardly, face to face, forming a 2 3/16 inch (5.6cm) wide four ply tab. Stitch the two lengthwise sides of tab 1/16 to 1/8 inch (1.6 to 3.2mm) from folded edges.	301	Similar to EFp-2	8-12	IB IB	IB IAA
	d. Turn one end of tab under approximately 3/8 inch (10mm) and place one end of the hook fastener tape over the raw edge and even with folded edge of tab end. Stitch all four sides of tape to tab 1/16 to 1/8 inch (1.6 to 3.2mm) from edges of tape.	301	EFa and LSbj-1	8-12	IB	IB
9.	Make waistband adjustment tabs.- a. Cut two strips of basic material approximately 3 3/4 inches (9.5cm) wide by 12 3/4 inches (32.4cm) long.					
	b. Cut one strip of 3/4 inch (19.0mm) wide elastic webbing 6 inches (15.2cm) long.					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
9.	Make waistband adjustment tabs.- (cont'd)					
	c. Fold the 12 3/4 inch (32.4cm) long strips for waistband tabs in half lengthwise, back to back, and turn under the raw edges as indicated by marks on pattern. Stitch folded edges 1/16 to 3/16 inch (1.6 to 4.8mm) from edge. The strip shall finish 1 1/16 to 1 3/16 inches (2.7 to 3.0cm) wide. The ends shall not be stitched closed.	301	EFp-2	8-12	IB	IB
	d. Insert one end of the elastic webbing approximately 1 inch (2.5cm) into one end of each waistband length. Secure by stitching with a 5/8 inch (15.9mm) cross boxstitch.	301	Cross-Boxstitch	8-12	IB	IB
	e. Cut two 3 3/8 to 3 5/8 inches (8.6 to 9.2cm) lengths of 1 inch (2.5cm) wide hook fastener tape.					
	f. Fold under the raw end of each waistband adjustment tab approximately 3/8 inch (6.4mm). Position the hook fastener tape over the folded under raw edge and stitch all sides of the tape 1/16 to 1/8 inch (1.6 to 3.2mm) from the edges.	301	LSbj-1	8-12	IB	IB
10.	Make hood (coated fabric outershell).-					
	a. Stitch darts in hood side panels, as indicated by marks on pattern, tapering and backtacking each pointed end of the darts.	301	OSf-1	8-12	IB	IB
	b. Stitch, face to face, hood center panel to hood side panel 1/4 to 3/8 inch (6.4 to 9.5mm) from raw edge.	301 or 401	LSq-2(a)	8-12	IB	IB
	c. Turn and topstitch 1/16 to 1/8 inch from fold with topstitch on center panel.	301 or 401	LSq-2(b)	8-12	IB IB	IB IAA

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
10.	Make hood (coated fabric outershell) (cont'd) d. Turn under ends of hood cord channel, back to back, 1/2 to 3/4 inch (12.7 to 19.0mm) and stitch 3/16 to 5/16 inch (4.8 to 7.9mm) from folded edge.	301 or 401	EFa-1	8-12	IB IB	IB IAA
	e. Cut a 44 to 46 inch (112 to 117cm) length of draw-cord.					
	f. Fold the hood cord channel, back to back, lengthwise with the length of cord inserted between. Position the raw edges of the hood cord channel on the face side of the hood face opening, as indicated on the patterns, and stitch 3/16 to 5/16 inch (4.8 to 7.9mm) from raw edge.	301 or 401	SSav-2(a) (with cord)	8-12	IB IB	IB IAA
	g. Knot ends of cord.					
*11.	Make bi-swing upper back.- a. Fold the back panel along the outer pattern marks, face to face, from the shoulder down to the bottom edge of back. Stitch with a single row of stitching approximately 1/16 inch (1.6mm) from the folded edge.	301 or 401	OSf-1	8-12	IB IB	IB IAA
	b. Fold the back panel along the inner pattern marks, with the face side out, and stitch with a double row of 1/4 inch (6.4mm) stitching. The outer row of stitching shall be approximately 1/16 inch (1.6mm) from the folded edge.	301 or 401	OSf-2	8-12	IB IB	IB IAA
	c. To complete bi-swing back, stay stitch across the pleat at top and bottom edges.					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Prepare front pockets and pocket welts.-					
	a. Fold pocket welt pieces lengthwise, back to back, as indicated by marks on pattern, with the non-woven interlining inserted between the plies and stitch the back raw edge along length of welt.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	b. Die cut 1/4 inch (6.4mm) diameter drainage holes in inner and outer pocket plies as indicated by pattern marks.					
	c. Stitch together the inner and outer pocket pieces, face to face, at the sides and bottom, 1/4 to 1/2 inch (6.4 to 12.7mm) from the raw edge.	301 or 401	SSa-1	8-12	IB IB	IB IAA
13.	Join pocket and welt to front.-					
	a. Place bottom welt piece on face side of front on front edge of pocket opening with the inner pocket piece on top of welt as indicated by pattern marks. Stitch 1/2 to 5/8 inch (12.7 to 15.9mm) from the raw edge.	301 or 401	Similar to SSbf-3(b)	8-12	IB IB	IB IAA
	b. Cut the front, as indicated by pattern marks, tongue notching each end. Turn pocket and bottom welt piece through the opening and topstitch 1/16 to 1/8 inch (1.6 to 3.2mm) from folded edge, taking care not to stitch through inner pocket piece.	301	Similar to SSbf-3(c)	8-12	IB	IB
	c. Turn under the raw edges of the pocket opening (tongues and facing) so that the opening is 3/4 to 7/8 inch (19.0 to 22.2mm) wide. Place top welt into finished position and stitch the 3 sides (top, back and bottom) 1/16 to 1/8 inch (1.6 to 3.2mm) from folded edge of front catching front, welt, and pocketing.	301	Similar to LSaj-1	8-12	IB	IB

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
13.	Join pocket and welt to front (cont'd) NOTE: The welts shall finish 3/4 to 7/8 inch (19.0 to 22.2mm) wide. The top welt shall be positioned so that the pocket opening faces the front of the coverall					
14.	Prepare fronts and trouser backs.-					
	a. Stitch darts in legs of front and in legs and waist of backs as indicated by marks on pattern, tapering and backtacking each pointed end of the darts.	301	OSF-1	8-12	IB	IB
	b. Die cut 1/4 inch (6.4mm) diameter holes in trouser legs as indicated by pattern marks.					
	c. Join leg adjustment tab to front as indicated by marks on pattern, stitching 3/16 to 5/16 inch (4.8 to 7.9mm) from unfinished end of tab.	301 or 401	LSb1-2(a)	8-12	IB IB	IB IAA
	d. Fold tab back over stitchline and secure back end of tab to leg with a 1 to 1 1/2 inches (2.5 to 3.8cm) long by full width of tab boxstitch. The top and bottom rows of stitching shall be superimposed on existing lines of stitching. Back row of stitching shall be 1/16 to 1/8 inch (1.6 to 3.2mm) from back folded edge.	301	Boxstitch	8-12	IB	IB
	e. Stitch backs of trousers together, face to face, at seat seam.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	f. Cut two 7 1/2 to 7 3/4 inch (19.0 to 19.7cm) long, 1 inch (2.5cm) wide pile fastener tapes. Join to fronts as indicated by pattern marks, stitching all four sides, 1/16 inch (1.6mm) from the edge, catching tops of hanging pockets.	301 or 401	LSbj-1	8-12	IB IB	IB IAA

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
15.	Join upper back to trouser backs.- Stitch the upper back to the trouser backs at the waist, face to face.	301 or 401	SSa-1	8-12	IB IB	IB IAA
16.	Join waistband and back waist channel to back.- a. Tack or bartack, 5/8 to 3/4 inch (15.9 to 19.0mm) long, the waistband (fastener tape side down) to the back with the waistband centered on the center seat seam and the waist seam.	301 or Bartack		8-12 21-28 per bartack	IB IB	IB IB
	b. Turn under the ends of the back waist channel approximately 5/8 inch (15.9mm) and topstitch 3/8 to 1/2 inch (9.5 to 12.7mm) from the folded edges.	301 or 401	EFa-1	8-12	IB IB	IB IAA
	c. Turn under the raw side edges of back waist channel 3/8 inch (9.5mm) and double stitch each folded edge to the back. The back waist channel shall be centered on the seat and waist seams and shall finish 2 3/8 to 2 5/8 inches (6.0 to 6.7cm) wide.	301 or 401	SSat-4	8-12	IB IB	IB IAA
*17.	Join shoulders.- Join the back to the fronts, face to face, at the shoulders.	301 or 401	SSa-1	8-12	IB IB	IB IAA
18.	Join accessory/utility pocket to the left sleeve.- Turn under all raw edges of the accessory/utility pocket 1/4 to 5/16 inch (6.4 to 7.9mm). Position on topsleeve, as indicated by marks on pattern and join to left sleeve with a double row of 1/16 to 1/8 inch (1.6 to 3.2mm) gauge stitching. NOTE: The pocket shall finish 4 5/8 to 4 7/8 inches (11.8 to 12.4cm) wide by 7 3/4 to 8 1/4 inches (20.0 to 21.0cm) long.	301 or 401	LSd-2	8-12	IB IB	IB IAA

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
19.	Join top and under sleeves and attach sleeve adjustment tapes.-					
	a. Join topsleeve and undersleeve, face to face, at sleeve outseam.	301 or 401	SSa-1	8-12	IB IB	IB IAA
	b. Join adjustment tab to topsleeve, as indicated by marks on pattern, by stitching 3/16 to 5/16 inch (4.8 to 7.9mm) from unfinished end of tab. Finished end of tab shall point towards back of coverall.	301 or 401	LSb1-2(a)	8-12	IB IB	IB IAA
	c. Fold adjustment tab back over stitchline and secure back end of tab to sleeve with a 1 to 1 1/2 inches (2.5 to 3.8cm) long by full width of tab boxstitch. The top and bottom rows of stitching shall be superimposed on existing lines of stitching. Back row of stitching shall be 1/16 to 1/8 inch (1.6 to 3.2mm) from folded edge.	301	Boxstitch	8-12	IB	IB
	d. Stitch a 5 3/4 to 6 1/4 inches (14.6 to 15.9cm) long by 2 inch (5.1cm) wide pile fastener tape to the sleeve, as indicated by marks on pattern, stitching all four sides 1/16 inch (1.6mm) from edge.	301	LSbj-1	8-12	IB	IB
	e. Die cut 1/4 inch (6.4mm) diameter drainage holes at bottom of sleeve as indicated by marks on pattern.					
20.	Join sleeves to body.- Join sleeves to body, face to face, at armholes.	301 or 401	SSa-1	8-12	IB IB	IB IAA
21.	Join sleeve inseams, body, and leg outseams.- a. Stitch sleeve inseams, body, and leg outseams, face to face, in one continuous operation.	301 or 401	SSa-1	8-12	IB IB	IB IAA

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Table I - Construction of coveralls (contd')

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
21.	Join sleeve inseams, body, and leg outseams.- (cont'd)					
	b. Stitch a 9 1/4 to 9 3/4 inches (24.1 to 24.8cm) long by 2 inch (5.1cm) wide pile fastener tape to the legs, as indicated by marks on pattern. All four sides of tape shall be stitched 1/16 inch (1.6mm) from the edges.	301	LSbj-1	8-12	IB	IB
22.	Join crotch seam.-					
	Join fronts, face to face, at the crotch seam.	301 or 401	SSa-1	8-12	IB IB	IB IAA
23.	Join leg inseams.-					
	Join leg inseams, face to face, in one continuous operation.	301 or 401	SSa-1	8-12	IB IB	IB IAA
24.	Join hood to body.-					
	Join hood to the body, face to face, at neckline.	301 or 401	SSa-1	8-12	IB IB	IB IAA
25.	Join combined innershell to outershell.-					
	a. Join the hood innershell to the hood outershell face to face.	301 or 401	SSax-2(a) (with cord)	8-12	IB	IB
	b. Turn to finished position and topstitch, 3/16 to 1/4 inch (4.8 to 6.4mm) from the seamline.	301 or 401	SSax-2(b) (with cord)	8-12	IB	IB
	c. Place body innershell and body outershell into finished position (inner shell legs and arms into outer shell legs and arms). Hem the sleeve and leg bottoms as follows:					

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
25.	Join combined innershell to outershell.- (cont'd) (1) Turn under the raw edge of outershell 1 1/4 inches (3.2cm). (2) Place the bottom edges of innershell between the plies of outershell up to the fold line. (3) Turn under the raw edge of outershell approximately 1/4 inch (6.4mm) and stitch 1/16 to 1/8 inch (1.6 to 3.2mm) from folded edge through all plies.	301 or 401	SS1-1	8-12	IB IB	IB IAA
	d. Turn under the self facings of the innershell and the outer shell, as indicated by pattern marks. Position the slide fastener between the plies with all folded fabric and fastener edges aligned. Topstitch 1/2 to 3/4 inch (12.7 to 19.0mm) from the folded edges. Beginning at the top of the zipper, down the side, and around the crotch, finishing at the zipper top on the other side. NOTE: The top stop of the slide fastener shall finish not more than 1/4 inch (6.4mm) from the finished neck seam.	301	LSe-1	8-12	IB	IB
*26.	Make and attach the front closure protective flap.- a. Fold the protective flap in half lengthwise, face to face, and stitch the top edge only.	301	SSa-1	8-12	IB	IB
	b. Turn flap to finished position and place the foam interlining piece between the plies and stitch across the top folded edge down the full length of the back raw edge without catching the foam interlining and across the bottom raw edge, 1/8 to 3/16 inch (3.2 to 4.8mm) from the edge.	301	SSa-1	8-12	IB	IB

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Table I - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
*26.	Make and attach the front closure protective flap.- (cont'd)					
	c. Stitch the protective flap to the inside of the right front and hood, 1/16 to 1/8 inch (1.6 to 3.2mm) from the raw edge, with the line of stitching superimposed on the stitching which secures the zipper to the right front. The top edge of the protective flap shall finish 1/8 to 3/8 inch (3.2 to 9.5mm) above the open end of the hood cord channel.	301	Similar to LSa-1	8-12	IB	IB
	d. Secure the bottom portion of the protective flap in the crotch area with a 5/8 to 3/4 inch (15.9 to 19.1mm) horizontal bartack with the stitching superimposed on the stitching which secures the bottom of slide fastener. When necessary, trim excess from bottom of protective flap.	Bartack		21-28 per bartack	IB	IB
27.	Clean.-					
	a. Trim all ends of stitching and remove loose threads.					
	b. Remove all spot, stains, shade, and size tickets.					

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* 3.8 Measurements.- Measurements of finished coveralls shall be as shown in Table II. All measurements and tolerances are expressed in inches and centimeters.

Table II - Sizes and finished measurements

Size	Chest A	Leg Inseam B	Length C	Sleeve Inseam D
Small (34-36)	47 (119.4)	27 (68.6)	59 (149.9)	19 1/2 (49.5)
Medium (38-40)	51 (129.5)	28 (71.1)	61 (154.9)	20 1/2 (52.1)
Large (42-44)	55 (139.7)	29 (73.7)	63 (160.0)	21 (53.3)
X-Large (46-48)	59 (149.9)	30 (76.2)	65 (165.1)	21 1/2 (54.6)
Tolerance	± 1 (2.5)	± 3/4 (1.9)	± 1 (2.5)	± 3/4 (1.9)

NOTE: Refer to figure 1 for A, B, C, and D measurements.
The coverall shall be laid flat with the front slide fastener completely closed.

A. Measure across front at base of armholes from folded edge to folded edge. Double the value obtained.
B. Measure from crotch seam along inseam to bottom of hemmed leg.
C. Measure from intersection of shoulder and hood seams to bottom of hemmed leg.
D. Measure from base of armhole along inseam to bottom of sleeve.

3.9 Workmanship.- The finished coveralls shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection.- When required (see 6.2), the first article submitted in accordance with 3.2 shall be visually inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except were otherwise indicated hereinafter.

* 4.4.1 Componet and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. A certificate of compliance shall be furnished for the colorfastness to laundering requirements of 3.3.5.1 and 3.3.5.2.1 and the colorfastness to crocking requirements of 3.3.7 when tested in accordance with 4.4.1.3. In addition, the expanded plastic foam shall be tested for buoyancy as specified in 4.4.1.1 and MIL-P-12420, except that the sample unit submitted for testing shall be 10 square feet (0.9 sq.m.). Three determinations shall be performed per sample unit of foam and the results reported to the nearest 0.1 pound.

* 4.4.1.1 Buoyancy.- The buoyancy test shall be made with a sample of the sheet material measuring 12 by 12 inches (30.5 by 30.5cm) by thickness of material furnished. Securely attach a spring scale, accurate to 0.25 pound, in a position directly over the test tank. Suspend a weighted wire basket from the scale in such a manner that the basket can be weighed while completely submerged in water. Proceed as follows:

(a) Weigh the empty basket under water.

(b) Place test sample inside the basket and submerge it so that the top of the basket is at least 2 inches (5.1cm) below the surface of water. Allow sample to remain submerged for 24 hours.

(c) After 24 hour submergence period, weigh the wire basket with the sample inside while both are still under water.

(d) Compute buoyancy in pounds per cubic foot:

$$\text{Lbs/CuFt} = \frac{\text{Weight of empty basket in water - Weight of basket and sample in water}}{\text{Volume of foam material in cubic feet}}$$

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* 4.4.1.2 Seam strength.- The cement shall be tested for seam strength properties in accordance with method 5100 of FED-STD-191 and the following exceptions: Cemented abutted seams shall be prepared using 4 by 6 inch (10.2 by 15.2cm) test specimens. Abutted seams shall be cemented across the 4 inch (10.2cm) dimension. The fully cured cemented seams shall be centered between and parallel to 2 inch (5.1cm) long clamps. The clamps shall be 1 inch (2.5cm) apart at start of test. The cement shall be unacceptable if one or more test speciments fail to meet the seam strength requirement specified in 3.3.10. A certificate of compliance shall be submitted for the identification requirement.

* 4.4.1.3 Colorfastness of tape, thong, and drawcord.- The zipper tape and nylon thong shall show good colorfastness to laundering when tested in accordance with Test Method 5614 of FED-STD-191. The dyed polyester drawcord shall show colorfastness to crocking not numerically below Munsell Value 8.5 when tested in accordance with Test Method 5651 of FED-STD-191.

4.4.2 Examination of the end item.- Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one finished coverall. The lot size shall be expressed in units of coveralls.

* 4.4.2.1 Visual examination.- The coveralls shall be examined for defects in color, design, material, construction, workmanship, and marking and the defects classified in accordance with the list below:

Defect	Classification	
	Major	Minor
I MATERIAL DEFECT(S) AND DAMAGE(S)		
a. Any size hole, needle chew, cut, tear, smash, visible mend or burn:		
(1) On outside	X	
(2) On inside		X
b. Misweave, missing yarn, or broken yarn; more than 1/2 inch (12.7mm) in any direction and on outside only		X
c. Permanent fold, pleat or crease in cloth; when caught in seaming or stitching - on outside only	X	
II CLEANNESS		
a. Any spot or stain on outside		X
b. Thread ends not trimmed or loose threads not removed on major portion of coverall		X
c. Two or more shade or size markings not removed		X
d. Any marking exposed or readily visible on outside		X

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Defect	Classification Major	Minor
III COMPONENT(S) AND ASSEMBLY (unless otherwise classifed herein)		
a. Any component not as specified (type, size, color, etc.)	X	
b. Any component part or operation omitted	X	
c. Any component part tight, twisted, distorted, pleated or full		X
d. Any operation not as specified		X
IV CUTTING		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X	
V SHADED PART(S)		
a. On outside:		
1. Variation in shade between two or more parts (except parts which may be cut from ends)	X	
2. Variation in shade within a part (except parts which may be cut from ends)		X
3. Part cut from ends badly shaded		X
VI SEAM(S) AND STITCHING(S)		
a. Accuracy of seaming		
1. Part of garment caught in any unrelated stitching operation	X	
2. Seam puckered, twisted, or pleated (unless otherwise classified herein)	X	
3. End of seam or stitching when not caught in another seam or stitching backtacked less than 3/8 inch (9.5mm)		X
4. Ends of a continuous line of stitching overlapped less than 1/2 inch (12.7mm)		X
5. Thread breaks not repaired or stiching back of the break less than 1/2 inch (12.7mm)		X
6. Restitch seam not neatly accomplished		X
7. Any seam restitched damaging material to a degree where it may tear along the row of stitching	X	

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Defect	Classification Major	Minor
VI SEAM(S) AND STITCHING(S) (cont'd)		
b. Stitching margin		
1. Top or edge stitching seam too close to edge resulting in damage to material:		
(a) Affecting appearance		X
(b) Affecting serviceability	X	
2. Irregular, affecting appearance		X
3. Beyond margin specified or varies more than 1/16 inch (1.6mm) when no margin is specified (score only when condition exists on more than 1/2 the length of seam)		X
c. Raw edges		
1. On outside:		
(a) -up to 1/2 inch (12.7mm) inclusive		X
(b) -more than 1/2 inch (12.7mm)	X	
2. On inside - more than 1/2 inch (12.7mm)		X
d. Open seams (see NOTE:)		
1. One outside:		
(a) -up to 1/4 inch (6.4mm) inclusive		X
(b) -more than 1/4 inch (6.4 mm)	X	
2. On inside:		
(a) -up to 1/2 inch (12.7mm) inclusive		X
(b) -more than 1/2 inch (12.7mm)	X	
NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam.		
e. Seam and stitch types		
1. Any required line of stitching omitted	X	
2. Not specified seam or stitch type	X	
3. Looper thread on outside or ends not caught in other seams or stitchings (when 401 sitich is used)		X
4. Any line of stitching not beginning or ending where specified		X

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Defect	Classification Major	Minor
VI SEAM(S) AND STITCHING(S) (cont'd)		
f. Stitch tension		
1. Loose tension resulting in loose seam	X	
2. Loose tension resulting in loosely exposed loops of lower or top thread (score only when condition exists on more than 1/2 the length of seam)		X
3. Tight tension (stitches break when normal strain is applied in the direction of seam or stitching)	X	
g. Stitches per inch (score only when condition exists on more than 1/2 the length of seam)		
1. One stitch less than the minimum or any number in excess of the maximum specified		X
2. Two or more stitches less than the minimum specified	X	
h. Bartack(s)		
1. Missing, insecure, misplaced or not serving intended purpose	X	
2. Not size specified		X
3. Loose stitch tensions		X
VII LABEL		
a. Missing, incorrect or illegible	X	
b. Misplaced		X
VIII LINING		
a. Too full or too short	X	
b. Twisted or pleated, affecting outside appearance		X
IX SLEEVE(S)		
Improperly set, resulting in excessive puckering around armhole		X
X UTILITY/ACCESSORY POCKET		
a. Missing	X	
b. Attached to right sleeve instead of left sleeve		X

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Defect	Classification Major	Minor
X UTILITY/ACCESSORY POCKET (cont'd)		
c. Slide fastener opening not facing toward front of coveralls	X	
d. Utility compartments not an integral part of combination pocket		X
e. One utility compartment missing		X
f. Two or more utility compartments missing	X	
g. Accessory pocket not formed with bellows		X
XI FACINGS (INSIDE AND OUTSIDE)		
a. Twisted, puckered, causing excessive fullness		X
b. Facings do not lay flat in cover slide fastener when slide fastener is closed		X
XII PROTECTIVE FLAP		
a. Twisted, tight or full; not permitting flap to lie smooth	X	
b. Top of flap less than 1/8 or more than 3/8 inch above hood cord channel opening		X
XIII ADJUSTMENT TABS		
Loop and pile fastener tapes not properly aligned, affecting performance		X
XIV HOOD		
a. Not topstitches as specified		X
b. Channel not attached as specified		X
c. Tie cord not knotted		X

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Defect	Classification Major	Minor
XV WAISTBAND AND BACK WAIST CHANNEL		
a. Elastic portion missing	X	
b. Hook fastener tape not on inside of waistband	X	
c. Channel not as specified		X
d. Channel not centered over waist seam		X
e. Finished width of waistband or back waist channel not as specified		X
XVI WELT POCKETS		
a. Welt poorly formed or irregular in width		X
b. Welt twisted, puckered or distorted		X
c. Pocket opening does not face towards front of coverall		X
d. Welts finish less than 3/4 inch (19.1mm) or more than 7/8 inch (22.2mm) wide		X
XVII FRONT SLIDE FASTENER		
a. Any part of assembly omitted, bent, broken, or otherwise defective	X	
b. Tape set on too loose or too tight, causing bulge, twist or fullness when closed	X	
c. Stitches too close to metal edge not permitting slider to pass freely	X	
d. Top stop set more than 1/4 inch (6.4mm) from finished neck seam	X	
e. Ends of slide fastener tape not securely caught in stitching	X	

NOTE: Slide fastener shall be checked for proper function by opening and closing slide fastener at least three times.

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Defect	Classification Major	Minor
XVIII DRAINAGE HOLES		
a. Any missing on pockets, sleeve or leg bottoms	X	
b. Not size specified		X
c. Not specified distance from sleeve or leg bottom	X	

4.4.2.2 Dimensional examination.- The coveralls shall be examined for defects in dimensions in accordance with the following:

a. Any measurement deviating from nominal dimensions and tolerances specified in Table II shall be scored as a measurement defect.

b. Sleeve lengths unequal in length by 1/2 inch (12.7mm) or more shall be scored as a size measurement defect.

4.4.2.3. Inspection levels and acceptable quality levels.- The inspection levels and acceptable quality levels (AQL's) expressed as defects per 100 units, for visual and dimensional examinations shall be as follows:

	AQL's	Inspection level
For visual examination in 4.4.2.1		
Major	2.5	II
Total (Major and Minor combined)	15.0	II
For dimensional examination in 4.4.2.2		
One class	4.0	S-3

* 4.4.2.4 In-process inspection.- Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material or operation for which in-process inspection had indicated nonconformance.

4.4.3 Examination of packaging requirements.- An examination shall be made to determine that packaging, packing and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closures of container flaps, improper taping, loose strapping, or inadequate stapling; bulging or distortion of container.
Content	Number of coveralls per shipping container is more or less than specified.

5. PACKAGING

5.1 Preservation-packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each pair of coveralls, with the slide fasteners closed, shall be laid flat, front down with sleeves and legs fully extended. Bring the legs up over the back approximately 25 inches (63.5cm). Fold the sleeves at right angles across the back. The length shall then be folded in half with the hood brought down over the chest. Each completely folded coverall shall measure approximately 23 by 15 inches (58 by 38cm). Each pair of folded coveralls shall be inserted in a close-fitting clear polyethylene film bag having a thickness of 0.003 inch (0.08mm) (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. A 1/4 inch (6.4mm) diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternately, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Level C.- Coveralls shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package may be the same as that normally used by the supplier for retail distribution when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Two coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced confoming to type SF, class weather-resistant, grade V2s, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A packages shall be packed flat, two in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

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* 5.2.2 Level B.- Two coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, grade 275, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A packages shall be packed flat, two in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2.1 Weather-resistant fiberboard container.- When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Level C (Commercial Packing).- Coveralls, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall comply with the U. S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethylene bagged packages.- Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

6.1 Intended use.- The coveralls are intended to be worn by Navy personnel as exposure suits on submarine decks to protect the wearer against any combination of wind, wet, and cold conditions.

6.2 Ordering date.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.2).
c. First article sample requirements (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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6.3 Samples and patterns.- For access to Samples and patterns address the procuring office issuing the invitation for bids.

6.4 Changes from previous issue.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Navy - SA

User activity:

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Preparing Activity:

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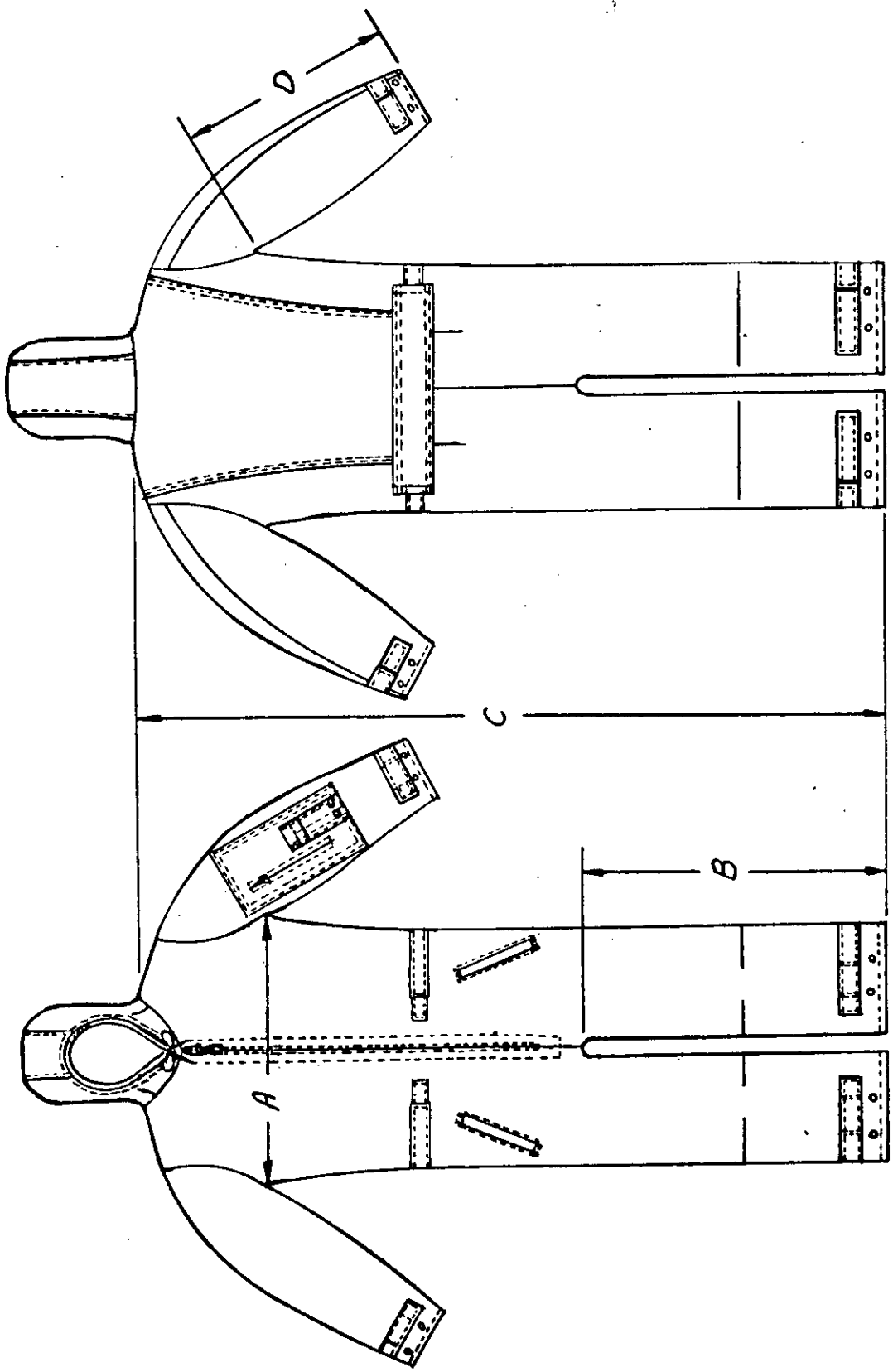


FIG-1 COVERALL, SUBMARINE DECK EXPOSURE

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